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Systems and methods for self-propelled garden machines

Abstract

Systems and methods which provide motive forces with respect to various apparatuses using self-engaging/disengaging transmission configurations are described. An autonomous-state transmission may comprise a transmission unit configured to autonomously engage and disengage transmission of power from a driving input member to a driven output member without utilization of external gearing or additional mechanical feedback. Embodiments of an autonomous-state transmission may utilize various roller-ramp clutch configurations in communication with override control for disengaging the roller-ramp clutch. The override control may implement various configurations of a skidding mechanism for facilitating engaging and/or disengaging the roller-ramp clutch. According to some examples, motion of the driving input member and/or driven output member is utilized for override force detection and providing override control with respect to the autonomous-state transmission. Embodiments of an autonomous-state transmission may be bidirectionally operative to autonomously engage and disengage transmission of power from a driving input to a driven output.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims the benefit of priority to U.S. patent application Ser. No. 18/560,031, filed on Nov. 9, 2023, which claims the benefit to PCT/CN2021/093340, filed on May 12, 2021, the disclosures of which are incorporated by reference herein in its entirety.

TECHNICAL FIELD

(2) The present invention relates generally to self-propelled garden machines and, more particularly, to self-engaging/disengaging transmission techniques with respect to self-propelled garden machines.

BACKGROUND OF THE INVENTION

(3) Garden machines of various forms are in common use for performing a number of routine tasks in daily life. For example, garden machines such as lawn mowers, snow throwers, grass edgers, tillers, etc. are used every day by persons performing work and leisure related tasks. These garden machines often comprise one or more motors of some form (e.g., electric motor or internal combustion motor) outputting power via a shaft which drive one or more implements configured for the related task. As examples, a lawn mower may drive an implement in the form of a grass cutting blade operable to shear grass, a snow thrower may drive implements in the form of an auger operable to gather and break down snow and impeller operable to propel the snow, a grass edger may drive an implement in the form of an edging blade operable to shear grass and trench soil, and a tiller may drive an implement in the form of a tine rotor operable to agitate soil. Using such

garden machines, a user may be substantially relieved of the labor required to manually perform the task. However, a user is nevertheless often required to supply motive forces with respect to the garden machine, such as to push and/or pull the garden machine within or throughout an area in which the related task is performed.

(4) Self-propelled garden machine configurations have become widespread in more recent times. A self-propelled garden machine may, for example, utilize one or more motor (e.g., electric motor or internal combustion motor) outputting power via a shaft to drive one or more wheels and/or other mobility members (e.g., tracks) for propelling the garden machine during use. A motor used for providing motive forces for a self-propelled garden machine may be the same or different motor than that used for powering the one or more implements. Irrespective of whether a separate motor is utilized for the motive forces, or a motor is shared between driving the implement(s) and providing the motive forces, a self-propelled garden machine can not only relieve the user of the manual labor for performing the task but may also relieve the user of much of the effort required to move the garden machine within or throughout an area for performing the task. Implementation of self-propelled operation of a garden machine is not, however, without challenges and disadvantages.

(5) The simplest of drivetrains (e.g., drivetrains having direct-drive, dog and spline, etc. configurations) implemented with respect to self-propelled garden machines generally prevent overrunning the motive operation of the garden machine. For example, a user is typically prevented from manually pushing the garden machine at a faster rate than provided by the self-propulsion implemented by the drivetrain when the garden machine is being self-propelled. Further, such simple drivetrains often present issues with respect to movement of the garden machine when the motor is stopped and/or moving the garden machine in a direction opposite the self-propulsion when under power by the drivetrain. For example, the wheels may remain engaged with the motor when the motor stops, requiring increased motive forces by the user (e.g., to overcome the friction of the wheels upon the ground surface and/or to manually turn the motor) if the garden machine is to be pushed or pulled in this state.

(6) More complex drivetrains (e.g., drivetrains having clutch-based transmission mechanisms) have been implemented in self-propelled garden machines to provide an improved user experience, as generally compared to that of a simple drivetrain implementation, with respect to motive operation of the garden machine. Such complex drivetrains, however, often come at the cost of configurations which are poorly suited for or are otherwise less than ideal in environments in which the garden machine may be used and fail to provide robust solutions for addressing a number of issues with respect to movement of a garden machine.

(7) As an example of a more complex drivetrain implementation, U.S. Pat. No. 10,524,417 to Fan et al., entitled "Walk-Behind, Self-Propelled Machine," (hereinafter "Fan", the disclosure of which is incorporated herein by reference) provides a clutch-based transmission mechanism for a lawn mower. The clutch-based transmission mechanism of Fan enables a user to manually push a lawn mower in the forward direction or pull the lawn mower in the reverse direction when the motor is turned off and the self-driving forward mode is exited and to likewise enable the user to manually push the lawn mower in the forward direction or pull the lawn mower in the reverse direction when the motor is turned off and the self-driving backward mode is exited. In providing this operability, the clutch-based transmission mechanism of Fan interfaces with the driven wheel using first and second transmission gears (e.g., gears disposed upon an output shaft of the clutch-based transmission) and corresponding first and second wheel gears (e.g., gears formed as part of the wheel assembly). Such an implementation, relying upon a combination of gears external to the clutch-based transmission mechanism itself and disposed at the wheel, is highly susceptible to damage and/or interference by lawn material (e.g., grass clippings, soil, stones, etc.) and is not well suited for the environment in which the lawn mower is used. Moreover, the clutch-based transmission mechanism implementation of Fan does not accommodate a user overrunning the

motive operation of the garden machine, and thus provides a configuration in which the user is prevented from pushing/pulling the lawn mower at a faster rate than provided by the self-propulsion when the motor is turned on and either the self-driving forward mode or self-driving backward mode is implemented. Similarly, the clutch-based transmission mechanism of Fan does not accommodate a user pushing/pulling the lawn mower in a direction opposite to the movement provided by the self-propulsion when the motor is turned on and either the self-driving forward mode or self-driving backward mode is implemented, and thus the user must provide increased motive forces to overcome that of the self-propelled drivetrain.

BRIEF SUMMARY OF THE INVENTION

(8) The present invention is directed to systems and methods which provide motive forces with respect to various apparatuses using self-engaging/disengaging transmission configurations. In accordance with embodiments of the invention, autonomous-state transmission implementations using self-engaging/disengaging transmission configurations are provided for self-propelled garden machine motive operation. For example, autonomous-state transmissions implemented in accordance with concepts of the present invention may be utilized in garden machines configured for various tasks, such as to provide robust self-propelled implementations of lawn mowers, snow throwers, grass edgers, tillers, fertilizer spreaders, garden carts, etc.

(9) An autonomous-state transmission of embodiments of the invention may comprise a transmission unit configured to autonomously control a force transfer state (e.g., engage and disengage) with respect to transmission of power from a driving input member (e.g., shaft, gear, etc.) to a driven output member (e.g., shaft, gear, etc.) without utilization of external gearing or additional mechanical feedback. For example, an autonomous-state transmission implemented according to some aspects of the present disclosure relies exclusively upon motion of the driving input member and/or driven output member for controlling a force transfer state thereof (e.g., engaging/disengaging transmission of power from the driving input member to the driven output member). According to some examples of an autonomous-state transmission, the motion of the driving input member and/or driven output member is utilized for override force (e.g., manual force sufficient to override self-propelled motive operation, manual pushing/pulling motive operation when a motor providing motive force is off, etc.) detection and providing override control with respect to the autonomous-state transmission. Accordingly, the complete mechanism of an autonomous-state transmission of embodiments of the invention may be wholly contained within a single enclosure or contiguous housing, such as may have only a driving input member and driven output member as external interfaces.

(10) Embodiments of an autonomous-state transmission may be bidirectionally operative to autonomously engage and disengage transmission of power from a driving input member to a driven output member. For example, a bidirectional autonomous-state transmission may operate to disengage (e.g., cease transmission of power from the driving input to the driven output) from an otherwise engaged state (e.g., providing motive forces for forward self-propulsion or backward self-propulsion) in response both to an overriding force in the motive direction (e.g., overrunning the motive operation) and an overriding force opposite the motive direction (e.g., underrunning the motive operation). As a specific example, a bidirectional autonomous-state transmission of an example embodiment is configured to transition from an engaged state to a disengaged state in response to a user pushing a walk-behind garden machine equipped with the bidirectional autonomous-state transmission with sufficient force to override self-propelled motive operation (e.g., override force) and manually push the garden machine at a faster rate than the self-propulsion provided by the bidirectional autonomous-state transmission (e.g., overrunning the bidirectional autonomous-state transmission). Further, the bidirectional autonomous-state transmission of this example embodiment is configured to transition from an engaged state to a disengaged state in response to the user pulling the walk-behind garden machine equipped with the bidirectional autonomous-state transmission with sufficient force to override self-propelled motive operation to

manually retard the movement of the garden machine at a slower rate than (or even reverse) the self-propulsion provided by the bidirectional autonomous-state transmission (e.g., underrunning the bidirectional autonomous-state transmission).

(11) Self-engaging/disengaging transmission configurations of an autonomous-state transmission may implement a clutch and associated override control. For example, embodiments of an autonomous-state transmission may utilize various roller-ramp clutch configurations in communication with override control for disengaging the roller-ramp clutch. The override control of the foregoing embodiments of an autonomous-state transmission may, for example, implement various configurations of a skidding mechanism for facilitating engaging and/or disengaging the roller-ramp clutch (e.g., based on override forces from overrunning and/or underrunning, manual motive operation, etc.). A skidding mechanism of embodiments of override control of an autonomous-state transmission may, for example, be configured for skidding action whereby the override control does not interfere with an engaged state of a clutch in normal motive operation. However, a skidding mechanism of embodiments of the override control may be configured for non-skidding action in response to override forces (e.g., overrunning and/or underrunning) transmitted via a driven output member to provide feedback control with respect to a disengaged state of the clutch.

(12) The foregoing has outlined rather broadly the features and technical advantages of the present invention in order that the detailed description of the invention that follows may be better understood. Additional features and advantages of the invention will be described hereinafter which form the subject of the claims of the invention. It should be appreciated by those skilled in the art that the conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present invention. It should also be realized by those skilled in the art that such equivalent constructions do not depart from the spirit and scope of the invention as set forth in the appended claims. The novel features which are believed to be characteristic of the invention, both as to its organization and method of operation, together with further objects and advantages will be better understood from the following description when considered in connection with the accompanying figures. It is to be expressly understood, however, that each of the figures is provided for the purpose of illustration and description only and is not intended as a definition of the limits of the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the present invention, reference is now made to the following descriptions taken in conjunction with the accompanying drawing, in which:
- (2) FIGS. 1A and 1B show examples of self-propelled garden machines in which autonomous-state transmissions of embodiments of the invention are disposed;
- (3) FIGS. 2A-2J show various views of a first exemplary autonomous-state transmission of embodiments of the invention;
- (4) FIGS. 3A-3F show various views of a second exemplary autonomous-state transmission of embodiments of the invention; and
- (5) FIGS. 4A-4D show schematic illustrations of operation of portions of a roller-ramp clutch mechanism utilized according to embodiments of the invention.

DETAILED DESCRIPTION OF THE INVENTION

(6) Various configurations of self-propelled garden machines may be used for performing a number of work or leisure related tasks. For example, garden machines such as lawn mowers, snow throwers, grass edgers, tillers, fertilizer spreaders, garden carts, etc. may be provided with one or more motors (e.g., electric motors or internal combustion motors) and associated drivetrains (e.g.,

driveshafts, gears, clutches, axles, etc.) configured for self-propelled motive operation of the garden machine. For example, one or more wheels and/or other mobility members (e.g., tracks) may be driven by one or more motors of a self-propelled garden machine so as to relieve a user of much of the effort required to move the garden machine, such as when operating the garden machine to perform a respective task.

(7) A drivetrain configuration of various garden machines may implement an autonomous-state transmission according to concepts of the present disclosure. In accordance with embodiments of the invention, autonomous-state transmission implementations comprise self-engaging/disengaging transmission configurations for self-propelled garden machine motive operation. For example, an autonomous-state transmission of embodiments of the invention may comprise a transmission unit configured to autonomously engage and disengage transmission of power from a driving input member (e.g., shaft, gear, etc.) to a driven output member (e.g., shaft, gear, etc.) without utilization of external gearing or additional mechanical feedback. Autonomous-state transmission configurations of embodiments of the invention may, for example, be utilized to facilitate robust self-propelled implementations of garden machines, such as by providing drivetrains that are well suited for use in environments in which the garden machine may be used, accommodating overrunning and/or underrunning of self-propelled motive operation, etc.

(8) FIGS. 1A and 1B show examples of self-propelled garden machines in which an autonomous-state transmission of embodiments of the invention is disposed. In particular, FIG. 1A illustrates an example of a self-propelled garden machine implementation as lawn mower **100A** and FIG. 1B illustrates an example of a self-propelled garden machine implementation as snow thrower **100B**. It should be appreciated that lawn mower **100A** and snow thrower **100B** are exemplary configurations of a self-propelled garden machine in which an autonomous-state transmission may be utilized and are not intended to limit the applicability of autonomous-state transmissions implemented in accordance with concepts herein.

(9) Each of the examples illustrated in FIGS. 1A and 1B includes autonomous-state transmission **110**, embodiments of which are further described below, configured according to concepts of the present invention to drive wheels **101**. Autonomous-state transmission **110** may, for example, be driven by a motor provided to supply motive power to the self-propelled garden machine. According to alternative embodiments, autonomous-state transmission **110** may be driven by a motor provided to drive an implement of the self-propelled garden machine.

(10) The complete mechanism of autonomous-state transmission **110** of the embodiments of FIGS. 1A and 1B is wholly contained within a single enclosure or contiguous housing. Accordingly, autonomous-state transmission **110** of the illustrated embodiments presents a configuration in which its external interfaces comprise only a driving input member and driven output member. Autonomous-state transmission **110** implemented according to some aspects of the present disclosure relies exclusively upon motion of the driving input member and/or driven output member for engaging/disengaging transmission of power from the driving input member to the driven output member.

(11) FIGS. 2A-2J and 3A-3F show details with respect to exemplary implementations of an autonomous-state transmission according to concepts of the present disclosure. Autonomous-state transmission **210** of FIGS. 2A-2J and autonomous-state transmission **310** of FIGS. 3A-3F may, for example, comprise a configuration of autonomous-state transmission **110** of FIG. 1A and/or FIG. 1B. It should be appreciated, however, that autonomous-state transmission **210** and autonomous-state transmission **310** may be utilized with respect to configurations of self-propelled garden machines in addition to or in the alternative to lawn mower **100A** and/or snow thrower **100B**.

(12) Referring first to FIGS. 2A-2J, the illustrated embodiment of autonomous-state transmission **210** will be described to aid in understanding concepts of the present invention. It should be understood that, although a particular configuration of autonomous-state transmission **210** is shown and described, various changes, substitutions, and alterations may be made with respect to one or

more components of the example of autonomous-state transmission **210** without departing from the spirit and scope of the invention.

(13) As shown in FIG. 2A, autonomous-state transmission **210** includes a driving input member (shown as driving shaft **201**) and driven output member (shown as driven shaft **202**). The driving input member may be provided motive power with respect to a self-propelled garden machine by one or more motors of the garden machine. Autonomous-state transmission **210** autonomously engages and disengages transmission of power from driving shaft **201** to driven shaft **202**. Driven shaft **202** may be in communication with one or more motive members (e.g., wheels, tracks, etc.) of a self-propelled garden machine so as to facilitate self-propelled motive operation of the garden machine. It should be appreciated that driving shaft **201** and/or driven shaft **202** of embodiments of autonomous-state transmission **210** may be configured differently than shown, such as to provide input and/or output gears, universal joints, etc.

(14) Autonomous-state transmission **210** of the illustrated embodiment in FIG. 2A is provided in a configuration in which its external interfaces comprise only a driving input member (driving shaft **201**) and a driven output member (driven shaft **202**), wherein the remainder of the componentry thereof is disposed within an enclosure or housing formed from mating cases **211** and **212**.

Accordingly, the motion of driving shaft **201** (e.g., the driving input) and/or driven shaft **202** (e.g., the driven output) of autonomous-state transmission **210** is utilized for override force (e.g., manual force sufficient to override self-propelled motive operation of the garden machine) detection and providing override control with respect to a force transfer state of autonomous-state transmission **210**.

(15) FIGS. 2B and 2C show autonomous-state transmission **210** having mating cases **211** and **212**, and corresponding input and output bearings, removed to thereby expose internal componentry thereof. Autonomous-state transmission **210** implements a clutch and associated override control configuration. Accordingly, the illustrated embodiment of autonomous-state transmission **210** includes clutch assembly **220** configured for selectively transferring motive force provided via driving shaft **201** to driven shaft **202**. Correspondingly, autonomous-state transmission **210** of the illustrated embodiment further includes override force feedback assembly **230**, comprising gears **231-234** and shaft **235**, configured for selectively communicating override forces provided via driven shaft **202** to clutch assembly **220**, facilitating override control with respect to autonomous-state transmission **210**.

(16) Clutch assembly **220** of embodiments of autonomous-state transmission **210** comprises a roller-ramp clutch configuration in communication with override force feedback assembly **230** facilitating control for disengaging the roller-ramp clutch. As shown in FIG. 2D, an exemplary roller-ramp clutch configuration of clutch assembly **220** comprises inner ring **225** coupled to driving shaft **201** and outer ring **227** coupled to driven shaft **202**. Inner ring **225** is preferably coupled to driving shaft **201** such that rotation of driving shaft is transferred to inner ring **225** substantially without slipping of inner ring with respect to driving shaft **201**. In contrast, override control interface **221** is preferably provided in a slipping relationship with respect to driving shaft **201**, such that at least some extent of rotational slippage of override control interface **221** is accommodated in relation to rotation of driving shaft **201**.

(17) In the roller-ramp clutch configuration of FIG. 2D, inner ring **225** nests within a circumference of inward surface **228** of outer ring **227**. Inner ring **225** comprises an eccentric outer surface comprising roller-ramp surfaces **226a-226f**, wherein the apexes provided at the junctions of the roller-ramp surfaces (e.g., the apex at the junction of roller-ramp surfaces **226a** and **226b**, the apex at the junction of roller-ramp surfaces **226b** and **226c**, and so on) are disposed more near inward surface **228** of outer ring **227** than is the remaining areas of the roller-ramp surfaces. Rolling members **224a-224f** (e.g., cylindrical pins) of clutch assembly **220** are disposed in the area between facing surfaces of a corresponding one of roller-ramp surfaces **226a-226f** and inward surface **228**. The space or gaps between the apexes of roller-ramp surfaces **226a-226f** and inward surface **228** is

preferably less than a diameter of rolling members **224a-224f**. Roller-ramp surfaces **226a-226f** of inner ring **225**, rolling members **224a-224f**, and inward surface **228** of outer ring **227** of embodiments cooperate to enable engaged and disengaged modes of clutch assembly **220**.

(18) FIGS. **4A-4D** illustrate cooperative interaction with respect to a roller-ramp surface of an inner ring, a rolling member, and an inward surface of an outer ring of an exemplary roller-ramp clutch configuration. Inner ring **425** may, for example, correspond to inner ring **225** of clutch assembly **220** described above and inner ring **325** of clutch assembly **320** described below, wherein roller-ramp surface **426** may represent any roller-ramp surface thereof. Similarly, rolling member **424** may represent any rolling member of clutch assembly **220** and clutch assembly **320**. Outer ring **427**, having inward surface **428**, may correspond to outer ring **227**, having inward surface **228**, and outer ring **327**, having inward surface **328**. As with the embodiments of clutch assemblies **220** and **320**, inner ring **425** may be coupled to a driving input member (e.g., driving shaft, gear, etc. in communication with a motor) and outer ring **427** may be coupled to a driven output member (e.g., driven shaft, gear, etc. in communication with a motive member).

(19) FIG. **4A** shows a disengaged mode with respect to the illustrated roller-ramp clutch configuration. For example, when the driving input member and driven output member are at rest (e.g., a motor for powering the driving input member is off and a garden machine in which the roller-ramp clutch configuration is disposed is not undergoing manual movement), rolling member **424** may rest at or towards the center of roller-ramp surface **426** (e.g., rolling member neutral or disengaged position) where the area between inward surface **428** and roller-ramp surface **426** is the greatest. Accordingly, rolling member **424** is not in simultaneous and/or firm contact with both roller-ramp surface **426** and inward surface **428**, and the roller-ramp clutch is disengaged. In the disengaged mode, outer ring **427** and inner ring **425** are free to move relative to one another (e.g., one ring may rotate while the other remains stationary, one ring may rotate at a different speed than the other ring, etc.). For example, as shown in the disengaged mode operation of FIG. **4B**, should a force be applied to cause outer ring **427** and inner ring **425** to move relative to one another (e.g., a garden machine in which the roller-ramp clutch configuration is disposed is undergoing manual movement from a user pushing the garden machine), rolling member **424** is free to roll in the area between facing surfaces of roller-ramp surface **426** and inward surface **428** (e.g., rolling member **424** disengaged position). In operation according to embodiments, outer ring **424** and inner ring **425** rotate together, whereby rolling member **424** is free to move within the space between the inner and outer rings (e.g., rolling member **424** may be encouraged to roll in one direction by outer ring **424** and in the opposite direction by roller-ramp surface **426** of inner ring **425** resulting in rolling member wandering within the space). Throughout this disengaged operation, rolling member **424** may remain within a central area of roller-ramp surface **426** (e.g., away from the apexes between roller-ramp surface **426** and adjacent roller-ramp surfaces of inner ring **425**), avoiding simultaneous and/or firm contact with both roller-ramp surface **426** and inward surface **428**, and the roller-ramp clutch configuration remains disengaged and does not transfer motive force between a driving input member and a driven output member.

(20) FIG. **4C** shows an engaged mode with respect to the illustrated roller-ramp clutch configuration. For example, when the driving input member is being driven (e.g., a motor for powering a garden machine in which the roller-ramp clutch configuration is disposed is applying force to the driving input) and inner ring **425** is rotated (e.g., clockwise in the illustration), rolling member **424** may be encouraged to move towards an end of roller-ramp surface **426** (e.g., near an apex between roller-ramp surface **426** and an adjacent roller-ramp surface of inner ring **425**) where the area between inward surface **428** and roller-ramp surface **426** is the greatest. Accordingly, rolling member **424** becomes simultaneously and firmly in contact with both roller-ramp surface **426** and inward surface **428** (e.g., pinched so as to prevent rolling of rolling member **424**, thereby disposing rolling member **424** in an engaged position), and the roller-ramp clutch is engaged. In the engaged mode, outer ring **427** and inner ring **425** are no longer free to move relative to one another.

Through this operation, the rolling-ramp clutch configuration in the engaged mode may transfer motive force between a driving input member and a driven output member.

(21) The engaged mode roller-ramp clutch of FIG. 4C may become disengaged in various ways. For example, an engaged roller-ramp clutch configuration may again become disengaged when application of force to inner ring 425 is ceased (e.g., motive force provided to a driving input by a motor for powering a garden machine in which the roller-ramp clutch configuration is disposed is terminated). As another example, an engaged roller-ramp clutch configuration may become disengaged when overrun (e.g., a sufficient overriding force is provided in the motive direction of a garden machine in which the roller-ramp clutch configuration is disposed overruns the motive operation of the garden machine). Disengaging of the roller-ramp clutch configuration due to overrunning is illustrated in FIG. 4D, wherein an overrunning force causes outer ring 427 to rotate at a more rapid rate than inner ring 425, at least for a short period of time (e.g., buffering operation of embodiments of the invention, described in further detail below, facilitates the outer ring briefly rotating at a more rapid rate than the inner ring followed by rotation of the other and inner rings remaining stationary relative to each other). Through this overrunning force, rolling member 424 is freed from its simultaneous and/or firm contact with both roller-ramp surface 426 and inward surface 428. Thus, rolling member 424 may migrate to a central area of roller-ramp surface 426 (e.g., away from the apexes between roller-ramp surface 426 and adjacent roller-ramp surfaces of inner ring 425), becoming disengaged and no longer transferring motive force between a driving input member and a driven output member.

(22) Referring again to FIG. 2D, the roller-ramp configuration of clutch assembly 220 engages to transfer motive force provided via driving shaft 201 to driven shaft 202 when one or more of rolling members 224a-224f are in simultaneous, firm contact (e.g., pinched so as to prevent their rolling) with their corresponding roller-ramp surface of inner ring 225 and inward surface 228 of outer ring 227. Accordingly, when driven shaft 202 is turned either clockwise or counterclockwise inner ring 225 turns accordingly, such that turning driven shaft 202 and inner ring 225 with sufficient speed and force will exert centrifugal forces sufficient to encourage rolling members 224a-224f toward respective apexes of the roller-ramp and engage inward surface 228 and be captured in a non-rolling relationship between outer ring 227 and inner ring 225. Inversely, the roller-ramp configuration of clutch assembly 220 disengages to preclude transfer of motive force from driving shaft 201 to driven shaft 202 when rolling members 224a-224f are in a rolling relationship with their corresponding roller-ramp surface of inner ring 225 and/or inward surface 228 of outer ring 227 (e.g., rolling within the area between a roller-ramp surface of roller-ramp surfaces 226a-226f and inward surface 228).

(23) The exemplary roller-ramp clutch configuration of clutch assembly 220 shown in FIG. 2D comprises override control interface 221 operative in cooperation with override force feedback assembly 230 for disengaging the roller-ramp clutch in response to override forces provided via driven shaft 202 to clutch assembly 220. Override control interface 221 of the illustrated embodiment includes gear portion 222 and rolling member holders 223a-223f. According to embodiments of clutch assembly 220, override control interface 221 is disposed in a partially nested relationship with respect to outer ring 227 and inner ring 225. A partially nested implementation of override control interface 221 disposes rolling member holders 223a-223f within the area between facing surfaces of inward surface 228 and roller-ramp surfaces 226a-226f, wherein rolling member holders 223a-223f are interleaved with rolling members 224a-224f. That is, in a partially nested implementation of embodiments, rolling member 224a is disposed between rolling member holders 223a and 223b, rolling member 224b is disposed between rolling member holders 223b and 223c, and so on including rolling member 224f disposed between rolling member holders 223f and 223a.

(24) In operation according to embodiments of the invention, rolling member holders 223a-223f of override control interface 221 may be utilized to encourage or otherwise facilitate engaged and/or

disengaged positioning of rolling members **224a-224f**. Rolling member holders **223a-223f** of embodiments may be controlled to facilitate engaged and/or disengaged positions with respect to rolling members **224a-224f** via force transferred by override force feedback assembly **230** through gear portion **222**. For example, an override force provided to driven shaft **202** may be transferred to override control interface **221** by override force feedback assembly **230** through gear portion **222**, which may cause rolling member holders **223a-223f** to move relative to rolling members **224a-224f** and encourage and/or discourage their engaged and/or disengaged positions. When an override force is not provided to driven shaft **202**, however, override force feedback assembly **230** is configured to enable override control interface **221** to facilitate rolling members **224a-224f** entering and/or remaining in engaged positions.

(25) FIG. 2E shows operation of autonomous-state transmission **210** with clutch assembly **220** in an engaged mode according to an example of the disclosure, wherein the direction of rotational force with respect to various components is indicated by the associated angular velocity vectors. The numerals provided in association with the angular velocity vectors shown in FIG. 2E designate the order of transmission of force by the gear components of the example autonomous-state transmission.

(26) The example of FIG. 2E may, for example, correspond to a situation in which a garden machine in which autonomous-state transmission **210** is disposed is operating in the self-propel mode. Although a particular direction (e.g., forward drive or rearward drive) of input/output rotational force (e.g., indicated by the clockwise and counterclockwise angular velocity vectors) is provided, the described operation applies to either forward drive or rearward drive (although the directions of rotational force are reversed).

(27) The example illustrated in FIG. 2E shows operation in which a drive force (e.g., 1.sup.st order clockwise input force) is applied to driving shaft **201**, whereby this force is transmitted to driven shaft **202** through clutch assembly **220** operating in an engaged mode to provide the drive force output (e.g., 2.sup.nd order clockwise output force). As shown in the cross-sectional view of clutch assembly **220** illustrated in FIG. 2F, rolling members **224a-224f** in their engaged positions may impinge upon rolling member holders **223a-223f** of override control interface **221**, whereby rolling member holders **223a-223f** operate to block rolling members **224a-224f** from moving from the engaged positions to disengaged positions. In this situation, override control interface **221** is encouraged to rotate according to the drive force applied to driving shaft **201**. Thus, gear portion **222** of override control interface **221** transmits force to gear **234** (e.g., inducing a 2.sup.nd order counterclockwise force) of override force feedback assembly **230**. Correspondingly, gear **234** transmits force to gear **233** (e.g., inducing a 3.sup.rd order clockwise force) of override force feedback assembly **230**. Simultaneously, gear **231** coupled to driven shaft **202** is encouraged to rotate clockwise (e.g., 2.sup.nd order clockwise output force) according to the drive force (e.g., 1.sup.st order clockwise input force) by drive members **229a-229d** of outer ring **227** engaging ends of circumferential slots **239a-239d** (FIG. 2D) disposed in gear **231**. Thus, gear **231** transmits force to gear **232** (e.g., inducing a 3.sup.rd order counterclockwise force) of override force feedback assembly **230**.

(28) As can be seen in the illustration of FIG. 2E, when operating in the exemplary engaged mode, gears **232** and **233** of override force feedback assembly **230** are rotating in contrariwise directions (e.g., gear **232** rotating in a counterclockwise direction and gear **233** rotating in a clockwise direction). Accordingly, shaft **235** of override force feedback assembly **230** preferably accommodates at least some extent of rotational slippage with respect to gears **232** and **233**. In accordance with some embodiments of the invention, shaft **235** and gear **233** are interfaced via a skidding mechanism facilitating limited slipping between shaft **235** and gear **233**.

(29) FIG. 2G shows an exploded diagram comprising an embodiment of gear **233** removed from shaft **235**, wherein portions of an exemplary skidding mechanism are visible. In the illustrated example, shaft **235** includes tab elements **236a** and **236b**. Tab elements **236a** and **236b** may, for

example, comprise spring biased sliding members disposed within a hub of shaft **235**. Correspondingly, in addition to having an orifice for slidably accepting the body of shaft **235**, gear **233** includes undulating inner surface **237**. Tab element **236a** and **236b** engage undulating inner surface **237** to facilitate skidding of gear **233** relative to rotation of shaft **235** when a suitable relative force is present with respect to the rotation of shaft **235** and gear **233**. Embodiments of the foregoing skidding mechanism are configured for skidding action whereby the override control does not interfere with an engaged state of a clutch in normal motive operation. For example, the depth of undulations of undulating inner surface **237**, the extended and/or contracted length of tab elements **236a** and **236b**, and/or the force of a spring bias provided to tab elements **236a** and **236b** encouraging their extension and protrusion from the hub of shaft **236** may be configured to accommodate skidding of gear **233** with respect to rotation of shaft **235** when gears **232** and **233** are rotating in contrariwise directions associated with autonomous-state transmission **210** with clutch assembly **220** in an engaged mode.

(30) FIG. 2H shows operation of autonomous-state transmission **210** with clutch assembly **220** in a disengaged mode according to an example of the disclosure, wherein the direction of rotational force with respect to various components is indicated by the associated angular velocity vectors. As with the previous example, the numerals provided in association with the angular velocity vectors shown in FIG. 2H designate the order of transmission of force by the gear components of the example autonomous-state transmission.

(31) The example of FIG. 2H may, for example, correspond to a situation in which a garden machine in which autonomous-state transmission **210** is disposed is being manually pushed while a motor for providing motive force is turned off. Although a particular direction (e.g., pushing or pulling of a garden machine) of rotational force (e.g., indicated by the clockwise and counterclockwise angular velocity vectors) is provided, the described operation applies to movement in either direction (although the directions of rotational force are reversed). Further, the operational description of the example also applies to overrun mode (e.g., where a user overruns the motive operation of the garden machine), although in that situation the clockwise and counterclockwise denotation is a relative rather than absolute. Accordingly, the illustrated configuration of autonomous-state transmission **210** provides a bidirectional autonomous-state transmission configuration which may operate to disengage (e.g., cease transmission of power from the driving input to the driven output) from an otherwise engaged state (e.g., providing motive forces for forward self-propulsion or backward self-propulsion) in response both to an overriding force in the motive direction (e.g., overrunning force associated with overrunning the motive operation) and an overriding force opposite the motive direction (e.g., underrunning force associated with underrunning the motive operation). In particular, bidirectional autonomous-state transmission **210** is configured to transition from an engaged state to a disengaged state in response to a user pushing a walk-behind garden machine equipped with the bidirectional autonomous-state transmission with sufficient force to override self-propelled motive operation (e.g., override force) and manually push the garden machine at a faster rate than the self-propulsion provided by the bidirectional autonomous-state transmission (e.g., overrunning the bidirectional autonomous-state transmission). Further, bidirectional autonomous-state transmission **210** is configured to transition from an engaged state to a disengaged state in response to the user pulling the walk-behind garden machine equipped with the bidirectional autonomous-state transmission with sufficient force to override self-propelled motive operation to manually retard the movement of the garden machine at a slower rate than (or even reverse) the self-propulsion provided by the bidirectional autonomous-state transmission (e.g., underrunning the bidirectional autonomous-state transmission).

(32) The example illustrated in FIG. 2H shows operation in which a force (e.g., 1.sup.st order counterclockwise force), such as associated with a user manually pushing or overrunning self-propelled operation of a garden machine, is applied to driven shaft **202**. In a situation where clutch assembly **220** was operating in an engaged mode immediately prior to the application of the noted

force at driven shaft **202**, rotation of shaft **202** may provide corresponding rotation of gear **231** (e.g., the 1.sup.st order counterclockwise force) attached thereto, without initially also causing rotation of outer ring **227** of clutch assembly **220**. That is, drive members **229a-229d** of outer ring **227** may be disposed to engage first ends of circumferential slots **239a-239d** (FIG. 2D) disposed in gear **231**, whereby rotation of shaft **202** and gear **231** proceeds without inducing rotation of outer ring **227** until drive members **229a-229d** traverses the lengths of circumferential slots **239a-239d** and engage second (opposite) ends thereof. Nevertheless, force is transmitted by gear **231** to gear **232** (e.g., inducing a 2.sup.nd order clockwise force) of override force feedback assembly **230**. Gear **232**, transmits the rotational force to shaft **235** attached thereto.

(33) As described above, shaft **235** of embodiments of override force feedback assembly **230** accommodates at least some extent of rotational slippage with respect to gears **232** and **233**. For example, the example illustrated in FIG. 2G provides a skidding mechanism facilitating limited slipping between shaft **235** and gear **233**. In the current example, a motor providing motive force is off, such that rotational force is not being transmitted to override force feedback assembly **230** by gear portion **222** of override control interface **221**. Similarly, in overrun mode operation, the relative rotational forces produce the same effect. Accordingly, in these situations, gear **233** of embodiments engages shaft **235** in a non-skidding relationship such that gear **232** transmits rotational force to gear **233** (e.g., inducing a 3.sup.rd order clockwise force). That is, although a hub of shaft **235** is configured to accommodate skidding of gear **233** with respect to rotation of shaft **235** when gears **232** and **233** are rotating in contrariwise directions associated with autonomous-state transmission **210** with clutch assembly **220** in an engaged mode, the hub is configured to engage gear **233** without skidding with respect to rotation of shaft **235** when autonomous-state transmission **210** is receiving override force via shaft **202**. Embodiments of the foregoing skidding mechanism are configured for non-skidding action in response to override forces (e.g., overrunning and/or underrunning) transmitted via shaft **202** to provide feedback control with respect to a disengaged state of clutch assembly **220**. For example, the depth of undulations of undulating inner surface **237** (FIG. 2G), the extended and/or contracted length of tab elements **236a** and **236b** (FIG. 2G), and/or the force of a spring bias provided to tab elements **236a** and **236b** encouraging their extension and protrusion from the hub of shaft **236** may be configured to accommodate non-skidding of gear **233** with respect to rotation of shaft **235** when overriding force is present with respect to shaft **202** consistent with operation of clutch assembly **220** in a disengaged mode. In accordance with embodiments of the invention, one or more aspects of a skidding mechanism of autonomous-state transmission **210** are selected so that skidding action occurs during self-propelled motive operation of a garden machine in which the autonomous-state transmission is disposed, while non-skidding action occurs during overrunning and/or underrunning the motive operation of the garden machine.

(34) In operation according to the illustrated example, force is transmitted by gear **233** to gear **234** (e.g., inducing 4th order counterclockwise force) of override force feedback assembly **230**. Gear **234** interfaces with gear portion **222** of override control interface **221**, and thus transmits force (e.g., clockwise 5th order clockwise force) to override control interface **221** of clutch assembly **220**. Rotation of override control interface **221** provides for corresponding movement of rolling member holders **223a-223f** thereof. As shown in FIG. 2I, where clutch assembly **220** is initially in an engaged mode (e.g., FIG. 2F), rolling member holders **223a-223f** may rotate (e.g., clockwise in this example) to move away from corresponding ones of rolling members **224a-224f** (e.g., rolling member holders **223a-223f** may cease to impinge upon rolling members **224a-224f**) and encourage or otherwise facilitate disengaged positioning of rolling members **224a-224f**.

(35) In accordance with embodiments of clutch assembly **220**, with rolling member holders **223a-223f** not in a position to encourage engaged positioning of rolling members **224a-224f** (e.g., rolling member holders **223a-223f** not impinging upon rolling members **224a-224f** to encourage an engaged position and/or rolling member holders **223a-223f** oppositely impinging upon rolling

members **224a-224f** to encourage a disengaged position), the rolling members may be freed from simultaneous and/or firm contact with both a corresponding roller-ramp surface **226a-226f** and inward surface **228**. For example, upon sufficient rotation of gear **231** for drive members **229a-229d** of outer ring **227** to engage second ends of circumferential slots **239a-239d**, outer ring **227** will be induced to rotate (e.g., counterclockwise force). Inward surface **228** may thus induce rotation of rolling members **224a-224f** such that the rolling members migrate to a central area of their respective roller-ramp surfaces **226a-226f** as shown in FIG. 2J. In operation according to embodiments, the sequence of the transmission of force and the buffering (e.g., facilitating a short period of time in which outer ring **227** can rotate at a more rapid rate than inner ring **225** while drive members **229a-229d** of outer ring **227** traverse circumferential slots **239a-239d** disposed in gear **231**, such as in response to an overrunning force, followed by the inner and outer rings again remain stationary relative to each other when drive members **229a-229d** engage the other end of slots **239a-239d**) implemented by drive members **229a-229d** disposed in circumferential slots **239a-239d** facilitates a disengaged mode of clutch assembly **220** and avoids any premature engagement of the clutch assembly. Accordingly, rolling members **224a-224f** may be disposed in a disengaged position and clutch assembly **220** disengaged. Rolling member holders **223a-223f** of embodiments preferably remain relatively positioned so as to encourage rolling members **224a-224f** to remain in a disengaged position throughout application of the override force at shaft **202**.

(36) Having described an embodiment of autonomous-state transmission **210** according to some examples of the present invention, attention is now directed to FIGS. 3A-3F, wherein the illustrated embodiment of autonomous-state transmission **310** will be described. It should be understood that, although a particular configuration of autonomous-state transmission **310** is shown and described, various changes, substitutions, and alterations may be made with respect to one or more components of the example of autonomous-state transmission **310** without departing from the spirit and scope of the invention.

(37) As shown in FIG. 3A, autonomous-state transmission **310** includes a driving input member (shown as driving shaft **301**) and driven output member (shown as driven gear **302**). The driving input member may be provided motive power with respect to a self-propelled garden machine by one or more motors of the garden machine. Autonomous-state transmission **310** autonomously engages and disengages transmission of power from driving shaft **301** to driven gear **302**. Driven gear **302** may be in communication with one or more motive members (e.g., wheels, tracks, etc.) of a self-propelled garden machine so as to facilitate self-propelled motive operation of the garden machine. It should be appreciated that driving shaft **301** and/or driven gear **302** of embodiments of autonomous-state transmission **310** may be configured differently than shown, such as to provide input and/or output gears, universal joints, etc.

(38) Autonomous-state transmission **310** of the illustrated embodiment in FIG. 3A is provided in a configuration in which its external interfaces comprise only a driving input member (driving shaft **301**) and a driven output member (driven gear **302**), wherein the remainder of the componentry thereof is disposed within an enclosure or housing formed from cover **311** and outer ring **327**. Accordingly, the motion of driving shaft **301** (e.g., the driving input) and/or driven gear **302** (e.g., the driven output) of autonomous-state transmission **310** is utilized for override force (e.g., manual force sufficient to override self-propelled motive operation of the garden machine) detection and providing override control with respect to a force transfer state of autonomous-state transmission **310**.

(39) FIGS. 3B and 3C show autonomous-state transmission **310** in an exploded view having cover **311** removed to thereby expose internal componentry thereof. Autonomous-state transmission **310** implements a clutch and associated override control configuration. Accordingly, the illustrated embodiment of autonomous-state transmission **310** includes clutch assembly **320**, comprising override control interface **221**, rolling members **324a-324f**, inner ring **325**, and outer ring **327**, configured for selectively transferring motive force provided via driving shaft **301** to driven gear

302. Correspondingly, autonomous-state transmission **310** of the illustrated embodiment further includes override force feedback assembly **330**, comprising gears **331-333**, configured for selectively communicating override forces provided via driven gear **302** to clutch assembly **320**, facilitating override control with respect to autonomous-state transmission **310**.

(40) The illustrated embodiment of clutch assembly **320** comprises a roller-ramp clutch configuration in communication with override force feedback assembly **330** facilitating control for disengaging the roller-ramp clutch. The exemplary roller-ramp clutch configuration of clutch assembly **320** comprises inner ring **325** coupled to driving shaft **301** (e.g., via a double-D slot configuration allowing some level hysteresis with respect to movement of inner ring **325** induced by driving shaft **301**) and outer ring **327** coupled to driven gear **302**. Inner ring **325** is preferably coupled to driving shaft **301** such that rotation of driving shaft is transferred to inner ring **325**. Override control interface **321** is preferably provided in a relationship with respect to driving shaft **301** that facilitates skidding, such that at least some extent of rotational slippage of override control interface **321** is accommodated in relation to rotation of driving shaft **301**.

(41) Similar to the roller-ramp clutch configuration of FIG. 2D above, inner ring **325** nests within a circumference of inward surface **328** of outer ring **327**. Inner ring **325** comprises an eccentric outer surface comprising roller-ramp surfaces **326a-326f**, wherein the apexes provided at the junctions of the roller-ramp surfaces are disposed more near inward surface **328** of outer ring **327** than is the remaining areas of the roller-ramp surfaces. Rolling members **324a-324f** (e.g., cylindrical pins) of clutch assembly **320** are disposed in the area between facing surfaces of a corresponding one of roller-ramp surfaces **326a-326f** and inward surface **328**. The space or gaps between the apexes of roller-ramp surfaces **326a-326f** and inward surface **328** is preferably less than a diameter of rolling members **324a-324f**. Roller-ramp surfaces **326a-326f** of inner ring **325**, rolling members **324a-324f**, and inward surface **328** of outer ring **327** of embodiments cooperate to enable engaged and disengaged modes of clutch assembly **320**. Cooperative interaction with respect to roller-ramp surfaces **326a-326f** of inner ring **325**, rolling members **324a-324f**, and inward surface **328** of outer ring **327** of the exemplary roller-ramp clutch configuration of clutch assembly **320** may be as described above with reference to FIGS. 4A-4D.

(42) The roller-ramp configuration of clutch assembly **320** engages to transfer motive force provided via driving shaft **301** to driven gear **302** when one or more of rolling members **324a-324f** are in simultaneous, firm contact (e.g., pinched so as to prevent their rolling) with their corresponding roller-ramp surface of inner ring **325** and inward surface **328** of outer ring **327**. Accordingly, when driven shaft **302** is turned either clockwise or counterclockwise inner ring may **325** turn accordingly, such that turning driven shaft **302** and inner ring **325** with sufficient speed and force will exert centrifugal forces sufficient to encourage rolling members **324a-324f** toward respective apexes of the roller-ramp and engage inward surface **328** and be captured in a non-rolling relationship between outer ring **327** and inner ring **325**. Inversely, the roller-ramp configuration of clutch assembly **320** disengages to preclude transfer of motive force from driving shaft **301** to driven gear **302** when rolling members **324a-324f** are in a rolling relationship with their corresponding roller-ramp surface of inner ring **325** and/or inward surface **328** of outer ring **327** (e.g., rolling within the area between a roller-ramp surface of roller-ramp surfaces **326a-326f** and inward surface **328**).

(43) The exemplary roller-ramp clutch configuration of clutch assembly **320** shown in FIGS. 3B and 3C comprises override control interface **321** operative in cooperation with override force feedback assembly **330** for disengaging the roller-ramp clutch in response to override forces provided via driven gear **302** to clutch assembly **320**. Override control interface **321** of the illustrated embodiment includes hub portion **322** and rolling member holders **323a-323f** affixed to or formed as part of hub portion **322**. According to embodiments of clutch assembly **320**, override control interface **321** is disposed in a nested relationship with respect to outer ring **327** and inner ring **325**. A nested implementation of override control interface **321** disposes rolling member

holders **323a-323f** within the area between facing surfaces of inward surface **328** and roller-ramp surfaces **326a-326f**, wherein rolling member holders **323a-323f** are interleaved with rolling members **324a-324f**. That is, in a nested implementation of embodiments, rolling member **324a** is disposed between rolling member holders **323a** and **323b**, rolling member **324b** is disposed between rolling member holders **323b** and **323c**, and so on including rolling member **324f** disposed between rolling member holders **323f** and **323a**.

(44) In operation according to embodiments of the invention, rolling member holders **323a-323f** of override control interface **321** may be utilized to encourage or otherwise facilitate engaged and/or disengaged positioning of rolling members **324a-324f**. Rolling member holders **323a-323f** of embodiments may be controlled to facilitate engaged and/or disengaged positions with respect to rolling members **324a-324f** via force transferred by override force feedback assembly **330** through hub portion **322**. For example, an override force provided to driven gear **302** may be transferred to override control interface **321** by override force feedback assembly **330** through hub portion **322**, which may cause rolling member holders **323a-323f** to move relative to rolling members **324a-324f** and encourage and/or discourage their engaged and/or disengaged positions. When an override force is not provided to driven gear **302**, however, override force feedback assembly **330** is configured to enable override control interface **321** to facilitate rolling members **324a-324f** entering and/or remaining in engaged positions.

(45) FIG. 3D shows operation of autonomous-state transmission **310** with clutch assembly **320** in an engaged mode according to an example of the disclosure, wherein the direction of rotational force with respect to various components is indicated by the associated angular velocity vectors. The numerals provided in association with the angular velocity vectors shown in FIG. 3D designate the order of transmission of force by the gear components of the example autonomous-state transmission.

(46) The example of FIG. 3D may, for example, correspond to a situation in which a garden machine in which autonomous-state transmission **310** is disposed is operating in the self-propel mode. Although a particular direction (e.g., forward drive or rearward drive) of input/output rotational force (e.g., indicated by the clockwise and counterclockwise angular velocity vectors) is provided, the described operation applies to either forward drive or rearward drive (although the directions of rotational force are reversed).

(47) The example illustrated in FIG. 3D shows operation in which a drive force (e.g., 1.sup.st order clockwise input force) is applied to driving shaft **301**, whereby this force is transmitted to outer ring **327** and thus driven gear **302** through clutch assembly **320** operating in an engaged mode to provide the drive force output (e.g., 2.sup.nd order clockwise output force). Rotation of outer ring **327** comprises rotation of ring gear **331** (e.g., 2.sup.nd order clockwise force) of override force feedback assembly **330** defined within a portion of outer ring **327**. Gears **332a-332c** of override force feedback assembly **330** are disposed as planetary gears within ring gear **331** of the illustrated embodiment (e.g., disposed upon pins affixed to or formed as part of hub portion **322**).

Accordingly, rotation of ring gear **331** (e.g., 2.sup.nd order clockwise force) induces rotation of gears **332a-332c** (e.g., 3.sup.rd order clockwise force). Gear **333** of override force feedback assembly **330** is disposed as a sun gear with respect to planetary gears **332a-332c** of the example embodiment. Thus, rotation of gears **332a-332c** (e.g., 3.sup.rd order clockwise force) induces rotation of gear **333** (e.g., 4th order counterclockwise force).

(48) Override control interface **321** is encouraged to rotate according to the drive force applied to driving shaft **301**. For example, as shown in the exploded diagram of FIG. 3E in which an embodiment of override control interface **321** is shown removed from driving shaft **301**, override control interface **321** is coupled to driving shaft **301** via a double-D slot configuration allowing some level hysteresis with respect to movement of override control interface **321** with respect to driving shaft **301**. Accordingly, rolling member holders **323a-323f** are enabled to move at least to some extent with respect to rolling members **324a-324f**. Rolling members **324a-324f** of clutch

assembly **320** in their engaged positions may impinge upon rolling member holders **323a-323f** of override control interface **321**, whereby rolling member holders **323a-323f** operate to block rolling members **324a-324f** from moving from the engaged positions to disengaged positions (e.g., similar to the configuration illustrated in FIG. 2F). During continued engaged operation of clutch assembly **320**, hub portion **322** of override control interface **321** is induced by the force applied to driving shaft **301** (e.g., 1.sup.st order clockwise force) to rotate accordingly (e.g., 2.sup.nd order clockwise force).

(49) As can be seen in the illustration of FIG. 3D, when operating in the exemplary engaged mode, gear **333** of override force feedback assembly **330** and hub portion **322** of override control interface **321** are rotating in contrariwise directions (e.g., gear **333** rotating in a counterclockwise direction and hub portion **322** rotating in a clockwise direction). Accordingly, gear **333** of override force feedback assembly **330** and/or hub portion **322** of override control interface **321** preferably accommodate at least some extent of rotational slippage with respect to each other. In accordance with some embodiments of the invention, gear **333** and hub portion **322** are interfaced via a skidding mechanism facilitating limited slipping between gear **333** and hub portion **322**.

(50) FIG. 3E shows portions of an exemplary skidding mechanism as may be utilized with respect to override control interface **321** and override force feedback assembly **330**. In the illustrated example, bias elements **335a** and **335b** are provided with tab elements **336a** and **336b** respectively. Gear **333** of the illustrated example is adapted to retain bias elements **335a** and **335b** for engaging hub portion **322**. Bias elements **335a** and **335b** may, for example, comprise spring biased members disposing tab elements **336a** and **336b** in communication with a surface of hub portion **322**. Correspondingly, in addition to having an orifice for accepting the body of shaft **301**, hub portion **322** includes undulating outer surface **337**. Tab element **336a** and **336b** engage undulating inner surface **337** to facilitate skidding of hub portion **322**, and correspondingly override control interface **321**, relative to rotation of gear **333** when a suitable relative force is present with respect to the rotation of override control interface **321** and gear **322**. Embodiments of the foregoing skidding mechanism are configured for skidding action whereby the override control does not interfere with an engaged state of a clutch in normal motive operation. For example, the depth of undulations of undulating outer surface **337**, the length of tab elements **336a** and **336b**, and/or the force of a spring bias provided by bias elements **335a** and **335b** may be configured to accommodate skidding of hub portion **322** with respect to rotation of gear **333** when hub portion **322** and gear **333** are rotating in contrariwise directions associated with autonomous-state transmission **310** with clutch assembly **320** in an engaged mode.

(51) FIG. 3F shows operation of autonomous-state transmission **310** with clutch assembly **320** in a disengaged mode according to an example of the disclosure. As with the previous examples, the numerals provided in association with the angular velocity vectors shown in FIG. 3F designate the order of transmission of force by the gear components of the example autonomous-state transmission.

(52) The example of FIG. 3F may, for example, correspond to a situation in which a garden machine in which autonomous-state transmission **310** is disposed is being manually pushed while a motor for providing motive force is turned off. Although a particular direction (e.g., pushing or pulling of a garden machine) of rotational force (e.g., indicated by the clockwise and counterclockwise angular velocity vectors) is provided, the described operation applies to movement in either direction (although the directions of rotational force are reversed). Further, the operational description of the example also applies to overrun mode (e.g., where a user overruns the motive operation of the garden machine), although in that situation the clockwise and counterclockwise denotation is a relative rather than absolute. Accordingly, the illustrated configuration of autonomous-state transmission **310** provides a bidirectional autonomous-state transmission configuration which may operate to disengage (e.g., cease transmission of power from the driving input to the driven output) from an otherwise engaged state (e.g., providing motive

forces for forward self-propulsion or backward self-propulsion) in response both to an overriding force in the motive direction (e.g., overrunning force associated with overrunning the motive operation) and an overriding force opposite the motive direction (e.g., underrunning force associated with underrunning the motive operation). In particular, bidirectional autonomous-state transmission **310** is configured to transition from an engaged state to a disengaged state in response to a user pushing a walk-behind garden machine equipped with the bidirectional autonomous-state transmission with sufficient force to override self-propelled motive operation (e.g., override force) and manually push the garden machine at a faster rate than the self-propulsion provided by the bidirectional autonomous-state transmission (e.g., overrunning the bidirectional autonomous-state transmission). Further, bidirectional autonomous-state transmission **310** is configured to transition from an engaged state to a disengaged state in response to the user pulling the walk-behind garden machine equipped with the bidirectional autonomous-state transmission with sufficient force to override self-propelled motive operation to manually retard the movement of the garden machine at a slower rate than (or even reverse) the self-propulsion provided by the bidirectional autonomous-state transmission (e.g., underrunning the bidirectional autonomous-state transmission).

(53) The example illustrated in FIG. **3F** shows operation in which a force (e.g., 1.sup.st order clockwise force), such as associated with a user manually pushing or overrunning self-propelled operation of a garden machine, is applied to driven gear **302**. In a situation where clutch assembly **320** was operating in an engaged mode immediately prior to the application of the noted force at driven gear **302**, rotation of gear **302** may provide corresponding rotation of outer ring **327** (e.g., the 1.sup.st order clockwise force) attached thereto. Rotation of outer ring **327** (e.g., 1.sup.st order clockwise force) correspondingly results in rotation of ring gear **331** (e.g., 1.sup.st order clockwise force) of override feedback assembly **330**. Gears **332a-332c** of override force feedback assembly **330** are disposed as planetary gears within ring gear **331** of the illustrated embodiment.

Accordingly, rotation of ring gear **331** (e.g., 1.sup.st order clockwise force) induces rotation of gears **332a-332c** (e.g., 2.sup.nd order clockwise force). Gear **333** of override force feedback assembly **330** is disposed as a sun gear with respect to planetary gears **332a-332c** of the example embodiment. Thus, rotation of gears **332a-332c** (e.g., 2.sup.nd order clockwise force) induces rotation of gear **333** (e.g., 3.sup.rd order counterclockwise force).

(54) As described above, gear **333** and hub portion **322** of embodiments accommodate at least some extent of rotational slippage with respect to gear **333** and hub portion **322**. For example, the example illustrated in FIG. **3E** provides a skidding mechanism facilitating limited slipping between gear **333** and hub portion **322**. In the current example, a motor providing motive force is off, such that rotational force is not being transmitted to override control interface **321**. Similarly, in overrun mode operation, the relative rotational forces produce the same effect. Accordingly, in these situations, hub portion **322** of embodiments engages gear **333** in a non-skidding relationship such that gear **333** transmits rotational force to hub portion **322** (e.g., inducing a 4.sup.th order clockwise force). That is, although hub portion **322** is configured to accommodate skidding of gear **333** with respect to rotation of gear **333** when gear **333** and hub portion **322** are rotating in contrariwise directions associated with autonomous-state transmission **310** with clutch assembly **320** in an engaged mode, hub portion **322** is configured to engage gear **333** without skidding with respect to rotation of gear **333** when autonomous-state transmission **310** is receiving override force via gear **302**. Embodiments of the foregoing skidding mechanism are configured for non-skidding action in response to override forces (e.g., overrunning and/or underrunning) transmitted via gear **302** to provide feedback control with respect to a disengaged state of clutch assembly **320**. For example, the depth of undulations of undulating inner surface **337** (FIG. **3E**), the length of tab elements **336a** and **336b**, and/or the force of a spring bias provided by bias elements **335a** and **335b** may be configured to accommodate non-skidding of hub portion **322** with respect to rotation of gear **333** when overriding force is present with respect to gear **302** consistent with operation of clutch assembly **320** in a disengaged mode. In accordance with embodiments of the invention, one

or more aspects of the skidding mechanism of autonomous-state transmission **310** are selected so that skidding action occurs during self-propelled motive operation of a garden machine in which the autonomous-state transmission is disposed, while non-skidding action occurs during overrunning and/or underrunning the motive operation of the garden machine.

(55) In operation according to the illustrated example, hub portion **322** provides force (e.g., 4.sup.th order clockwise force) to override control interface **321** of clutch assembly **320**. Rotation of override control interface **321** provides for corresponding movement of rolling member holders **323a-323f** thereof, without initially also causing rotation of inner ring **325** of clutch assembly **320**. That is, the double-D shaped circumference of orifice **329** of gear **333** coupling embodiments of override control interface **321** to driving shaft **301** allows some level hysteresis with respect to movement of override control interface **321** relative to driving shaft **301** and correspondingly inner ring **325**. Rotation of override control interface **321** proceeds without inducing rotation of inner ring **325** until the double-D circumference of orifice **329** is traversed by the flat surfaces **311a** and **311b** of driven shaft **301** and engage second (opposite) sides thereof (e.g., thereafter inducing 5.sup.th order counterclockwise rotation of driving shaft **301**). Where clutch assembly **320** is initially in an engaged mode (e.g., similar to the illustration of FIG. 2F), rolling member holders **323a-323f** may rotate to move away from corresponding ones of rolling members **324a-324f** (e.g., rolling member holders **323a-323f** may cease to impinge upon rolling members **324a-324f**, similar to the illustration of FIG. 2I) and encourage or otherwise facilitate disengaged positioning of rolling members **324a-324f**.

(56) Buffering resulting from interaction between the double-D circumference of orifice **329** of gear **333** interfacing with flat surfaces **311a** and **311b** of driving shaft **301** disposed within orifice **329** facilitates a disengaged mode of clutch assembly **320** according to some embodiments of the invention. In an example of disengaging of the roller-ramp clutch configuration, an overrunning force may cause outer ring **327** to rotate at a more rapid rate than inner ring **325**, at least for a short period of time. The sequence of the transmission of force and the buffering (e.g., facilitating a short period of time in which outer ring **327** can rotate at a more rapid rate than inner ring **325** while flat surfaces **311a** and **311b** of driving shaft **301** rotate within the double-D circumference of orifice **329**, such as in response to an overrunning force, followed by the inner and outer rings again remaining stationary relative to each other when flat surfaces **311a** and **311b** engage the other side of the double-D circumference of orifice **329**) implemented by flat surfaces **311a** and **311b** disposed in the double-D circumference of orifice **329** facilitates a disengaged mode of clutch assembly **320**. Accordingly, rolling members **324a-324f** may be disposed in a disengaged position and clutch assembly **320** disengaged. Rolling member holders **323a-323f** of embodiments preferably remain relatively positioned so as to encourage rolling members **324a-324f** to remain in a disengaged position throughout application of the override force at driven gear **302**.

(57) In accordance with embodiments of clutch assembly **320**, with rolling member holders **323a-323f** not in a position to encourage engaged positioning of rolling members **324a-324f** (e.g., rolling member holders **323a-323f** not impinging upon rolling members **324a-324f** to encourage an engaged position and/or rolling member holders **323a-323f** oppositely impinging upon rolling members **324a-324f** to encourage a disengaged position), the rolling members may be freed from simultaneous and/or firm contact with both a corresponding roller-ramp surface **326a-326f** and inward surface **328**. For example, upon sufficient rotation of hub portion **322**, inward surface **328** may induce rotation of rolling members **324a-324f** such that the rolling members migrate to a central area of their respective roller-ramp surfaces **326a-326f** (e.g., similar to the illustration of FIG. 2J). In operation according to embodiments, the sequence of the transmission of force facilitates a disengaged mode of clutch assembly **320**. Accordingly, rolling members **324a-324f** may be disposed in a disengaged position and clutch assembly **320** disengaged. Rolling member holders **323a-323f** of embodiments preferably remain relatively positioned so as to encourage rolling members **324a-324f** to remain in a disengaged position throughout application of the

override force at gear **302**.

(58) Although the examples above have been described with reference to a particular direction of applied force in order to provide details for an understanding of the concepts herein, it should be appreciated that autonomous-state transmissions of embodiments are nevertheless bidirectionally operative to autonomously engage and disengage transmission of power from a driving input member to a driven output member. Bidirectional autonomous-state transmissions of embodiments of the invention may, for example, operate to disengage from an otherwise engaged state (e.g., providing motive forces for forward self-propulsion or backward self-propulsion) in response both to an overriding force in the motive direction (e.g., overrunning the motive operation) and an overriding force opposite the motive direction (e.g., underrunning the motive operation).

(59) It can be seen from the examples above that autonomous-state transmissions of embodiments of the invention comprise a transmission unit configured to autonomously engage and disengage transmission of power from a driving input member to a driven output member without utilization of gearing or additional mechanical feedback external to the transmission unit which is wholly contained within a single enclosure or contiguous housing (e.g., having only a driving input and driven output as external interfaces). In operation of autonomous-state transmissions **210** and **310** according to examples herein, the autonomous-state transmissions may rely exclusively upon motion of the driving input member and/or driven output member for engaging/disengaging transmission of power from the driving input member to the driven output member.

(60) Although the present invention and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the invention as defined by the appended claims. Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure of the present invention, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present invention. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

(61) Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification.

Claims

1. An autonomous-state transmission for providing self-propelled motive operation of a garden machine, the autonomous-state transmission comprising a driving shaft; a driven shaft; a clutch assembly disposed in mechanical communication between the driving shaft and the driven shaft to selectively engage power transmission from the driving shaft to the driven shaft, the clutch assembly comprising an inner ring coupled to one of the driving shaft and the driven shaft, an outer ring coupled to the other of the driving shaft and the driven shaft, and an override control interface disposed on the driving shaft, the clutch assembly being movable between an engaged position and a disengaged position, the engaged position providing the inner ring and the outer ring as a rotatably coupled pair to rotate together, and the disengaged position providing the inner ring and the outer ring as a rotatably decoupled pair to permit relative rotation between the inner ring and the outer ring; and an override force feedback assembly in mechanical communication with the driven shaft and the override control interface to selectively communicate an override force from the driven shaft, the override force feedback (OFF) assembly comprising an OFF shaft spaced apart

from the driving shaft and the driven shaft, a first gear disposed on the driven shaft, a second gear disposed on the OFF shaft in mechanical communication with the first gear, and a third gear disposed on the OFF shaft in mechanical communication with the override control interface.

2. The autonomous-state transmission of claim 1, wherein the inner ring is fixed to the driving shaft to rotate therewith, and wherein the outer ring is fixed to the driven shaft to rotate therewith.

3. The autonomous-state transmission of claim 1, wherein the inner ring is nested within the outer ring.

4. The autonomous-state transmission of claim 1, wherein the inner ring comprises an eccentric outer surface comprising one or more roller-ramp surfaces.

5. The autonomous-state transmission of claim 4, wherein the override control interface further comprises one or more cylindrical pins disposed between the eccentric outer surface of the inner ring and an inward surface of the outer ring.

6. The autonomous-state transmission of claim 5, wherein the one or more cylindrical pins are each movable along a corresponding roller ramp surface of the one or more roller-ramp surfaces.

7. The autonomous-state transmission of claim 5, wherein the engaged position comprises the one or more cylindrical pins in simultaneous contact with the eccentric outer surface of the inner ring and the inward surface of the outer ring.

8. The autonomous-state transmission of claim 5, wherein the disengaged position comprises the one or more cylindrical pins in non-simultaneous contact with only one of the eccentric outer surface of the inner ring and the inward surface of the outer ring.

9. The autonomous-state transmission of claim 5, wherein the override control interface comprises a gear portion and a rolling member holder portion joined to the gear portion to rotate therewith, the one or more cylindrical pins disposed within the rolling member holder portion.

10. The autonomous-state transmission of claim 1, wherein the override control interface is coaxially coupled to the driving shaft in a slipping relationship to permit at least some extent of rotational slippage in relation to rotation of the driving shaft.

11. The autonomous-state transmission of claim 1, wherein the override control interface comprises a gear portion having a plurality of teeth, the clutch assembly further comprising a skidding mechanism comprising one or more biased tab elements disposed on an inner surface of the gear portion, and a hub disposed on the driving shaft and comprising the driving shaft and the gear portion, the hub comprising an undulating surface as an outer surface thereof, wherein the one or more biased tab elements are interfaced with the undulating surface.

12. The autonomous-state transmission of claim 1, wherein the third gear comprises an undulating inner surface, the OFF assembly further comprising a skidding mechanism disposed between the OFF shaft and the third gear, the skidding mechanism comprising a hub on the OFF shaft, the hub supporting one or more biased tab elements interfaced with the undulating inner surface of the third gear.

13. An autonomous-state transmission for providing self-propelled motive operation of a garden machine, the autonomous-state transmission comprising a driving shaft; a driven shaft; a clutch assembly disposed in mechanical communication between the driving shaft and the driven shaft to selectively engage power transmission from the driving shaft to the driven shaft, the clutch assembly comprising an inner ring coupled to one of the driving shaft and the driven shaft, the inner ring comprising an eccentric outer surface comprising one or more roller-ramp surfaces, an outer ring coupled to the other of the driving shaft and the driven shaft, the outer ring having the inner ring nested therein, and an override control interface comprising one or more cylindrical pins disposed between the eccentric outer surface of the inner ring and an inward surface of the outer ring, the clutch assembly being movable between an engaged position and a disengaged position, the engaged position comprising the one or more cylindrical pins in simultaneous contact with the eccentric outer surface of the inner ring and the inward surface of the outer ring, and the disengaged position comprising the one or more cylindrical pins in non-simultaneous contact with

only one of the eccentric outer surface of the inner ring and the inward surface of the outer ring; and an override force feedback assembly in mechanical communication with the driven shaft and the override control interface to selectively communicate an override force from the driven shaft, the override force feedback (OFF) assembly comprising an OFF shaft spaced apart from the driving shaft and the driven shaft, a first gear disposed on the driven shaft, a second gear disposed on the OFF shaft in mechanical communication with the first gear, and a third gear disposed on the OFF shaft in mechanical communication with the override control interface.

14. The autonomous-state transmission of claim 13, wherein the inner ring is fixed to the driving shaft to rotate therewith, and wherein the outer ring is fixed to the driven shaft to rotate therewith.

15. The autonomous-state transmission of claim 13, wherein the one or more cylindrical pins are each movable along a corresponding roller ramp surface of the one or more roller-ramp surfaces.

16. The autonomous-state transmission of claim 13, wherein the override control interface comprises a gear portion and a rolling member holder portion joined to the gear portion to rotate therewith, the one or more cylindrical pins disposed within the rolling member holder portion.

17. The autonomous-state transmission of claim 13, wherein the override control interface is coaxially coupled to the driving shaft in a slipping relationship to permit at least some extent of rotational slippage in relation to rotation of the driving shaft.

18. The autonomous-state transmission of claim 13, wherein the override control interface comprises a gear portion having a plurality of teeth, the clutch assembly further comprising a skidding mechanism comprising one or more biased tab elements disposed on an inner surface of the gear portion, and a hub disposed on the driving shaft and comprising the driving shaft and the gear portion, the hub comprising an undulating surface as an outer surface thereof, wherein the one or more biased tab elements are interfaced with the undulating surface.

19. The autonomous-state transmission of claim 13, wherein the third gear comprises an undulating inner surface, the OFF assembly further comprising a skidding mechanism disposed between the OFF shaft and the third gear, the skidding mechanism comprising a hub on the OFF shaft, the hub supporting one or more biased tab elements interfaced with the undulating inner surface of the third gear.
